

1

$$\log\left(\frac{p}{1-p}\right) = z = w^T x + b$$

Thus it follows that $p = \frac{1}{1+e^{-z}}$

So

$$p_1 = 0.6456563062, p_2 = 0.0003353501, p_3 = 0.9975273768, p_4 = 0.0000000021, p_5 = 0^*$$

* Note this is due to rounding as the probability in this case asymptotically approaches 0 and never reaches exact 0

Thus the one we are most confident on is z^5 because its probability is furthest from 0.5 and the one we are least confident on is z^1 as its probability is closes to 0.5. We could also look at it as the z 's distance from 0 where further from 0 is more certain and closer to 0 is less certain.

2

For the full derivation we simply replace the $\sigma(z^i)$ with $1 - \sigma(z^i)$ thus:

$$L = -\sum_{i=1}^N y^i \log(1 - \sigma(z^i)) + (1 - y^i) \log(\sigma(z^i))$$

For w_j :

$$\frac{\delta L}{\delta w_j} = \sum_{i=1}^N -y^i \frac{1}{1-\sigma(z^i)} (-\sigma(z^i))(1 - \sigma(z^i))x_j^i - (1 - y^i) \frac{1}{\sigma(z^i)} \sigma(z^i)(1 - \sigma(z^i))x_j^i$$

$$\frac{\delta L}{\delta w_j} = \sum_{i=1}^N y^i \sigma(z^i)x_j^i - (1 - y^i)(1 - \sigma(z^i))x_j^i$$

$$\frac{\delta L}{\delta w_j} = \sum_{i=1}^N y^i \sigma(z^i)x_j^i - (1 - y^i - \sigma(z^i) + y^i \sigma(z^i))x_j^i$$

$$\frac{\delta L}{\delta w_j} = \sum_{i=1}^N (y^i - (1 - \sigma(z^i)))x_j^i$$

For b :

$$\frac{\delta L}{\delta b} = \sum_{i=1}^N -y^i \frac{1}{1-\sigma(z^i)} (-\sigma(z^i))(1 - \sigma(z^i)) - (1 - y^i) \frac{1}{\sigma(z^i)} \sigma(z^i)(1 - \sigma(z^i))$$

$$\frac{\delta L}{\delta b} = \sum_{i=1}^N y^i \sigma(z^i) - (1 - y^i)(1 - \sigma(z^i))$$

$$\frac{\delta L}{\delta b} = \sum_{i=1}^N y^i \sigma(z^i) - (1 - y^i - \sigma(z^i) + y^i \sigma(z^i))$$

$$\frac{\delta L}{\delta b} = \sum_{i=1}^N (y^i - (1 - \sigma(z^i)))$$

Thus it follows that

$$w_j \leftarrow w_j - \eta \frac{\delta L}{\delta w_j}$$

$$b \leftarrow b - \eta \frac{\delta L}{\delta b}$$

3

$$L = \frac{\lambda}{2n} \sum_{i=1}^m w_i^2 - \sum_{i=1}^N y^i \log(\sigma(z^i)) + (1 - y^i) \log(1 - \sigma(z^i))$$

For w_j :

$$\frac{\delta L}{\delta w_j} = \frac{\lambda w_j}{n} + \sum_{i=1}^N -y^i \frac{1}{\sigma(z^i)} \sigma(z^i)(1 - \sigma(z^i)) x_j^i + (1 - y^i) \frac{1}{1 - \sigma(z^i)} \sigma(z^i)(1 - \sigma(z^i)) x_j^i$$

$$\frac{\delta L}{\delta w_j} = \sum_{i=1}^N -y^i (1 - \sigma(z^i)) x_j^i + (1 - y^i) (\sigma(z^i)) x_j^i + \frac{\lambda w_j}{n}$$

$$\frac{\delta L}{\delta w_j} = \sum_{i=1}^N (\sigma(z^i) - y^i) x_j^i + \frac{\lambda w_j}{n}$$

For b :

$$\frac{\delta L}{\delta w_j} = - \sum_{i=1}^N y^i \frac{1}{\sigma(z^i)} \sigma(z^i)(1 - \sigma(z^i)) x_j^i + (1 - y^i) \frac{1}{1 - \sigma(z^i)} \sigma(z^i)(1 - \sigma(z^i)) x_j^i$$

$$\frac{\delta L}{\delta w_j} = \sum_{i=1}^N -y^i (1 - \sigma(z^i)) x_j^i + (1 - y^i) (\sigma(z^i)) x_j^i$$

$$\frac{\delta L}{\delta w_j} = \sum_{i=1}^N (\sigma(z^i) - y^i) x_j^i$$

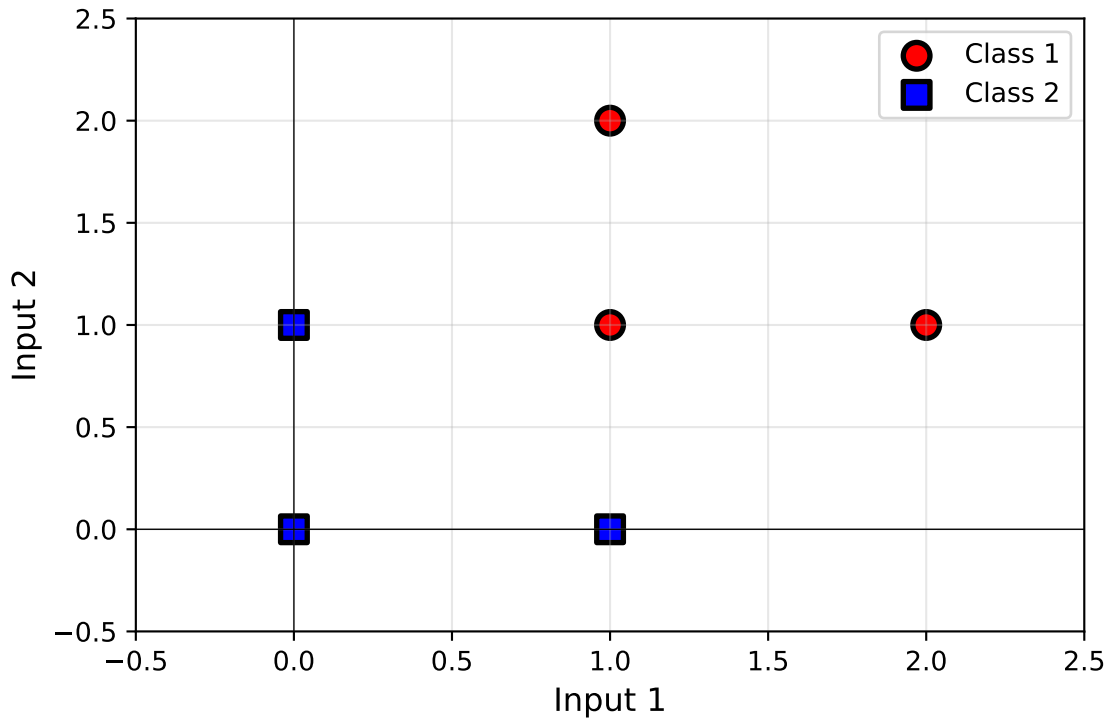
Again it still applies that

$$w_j \leftarrow w_j - \eta \frac{\delta L}{\delta w_j}$$

$$b \leftarrow b - \eta \frac{\delta L}{\delta b}$$

4

a



The dataset is linearly separable by the line $y = 1.5 - x$

b

Decision boundary: $2y + 2x - 3 = 0$

Hyperplane: $2x + 2y - 3 = \pm 1$

So class 1 hyperplane: $2x + 2y - 4 = 0$

So class 2 hyperplane: $2x + 2y - 2 = 0$

c

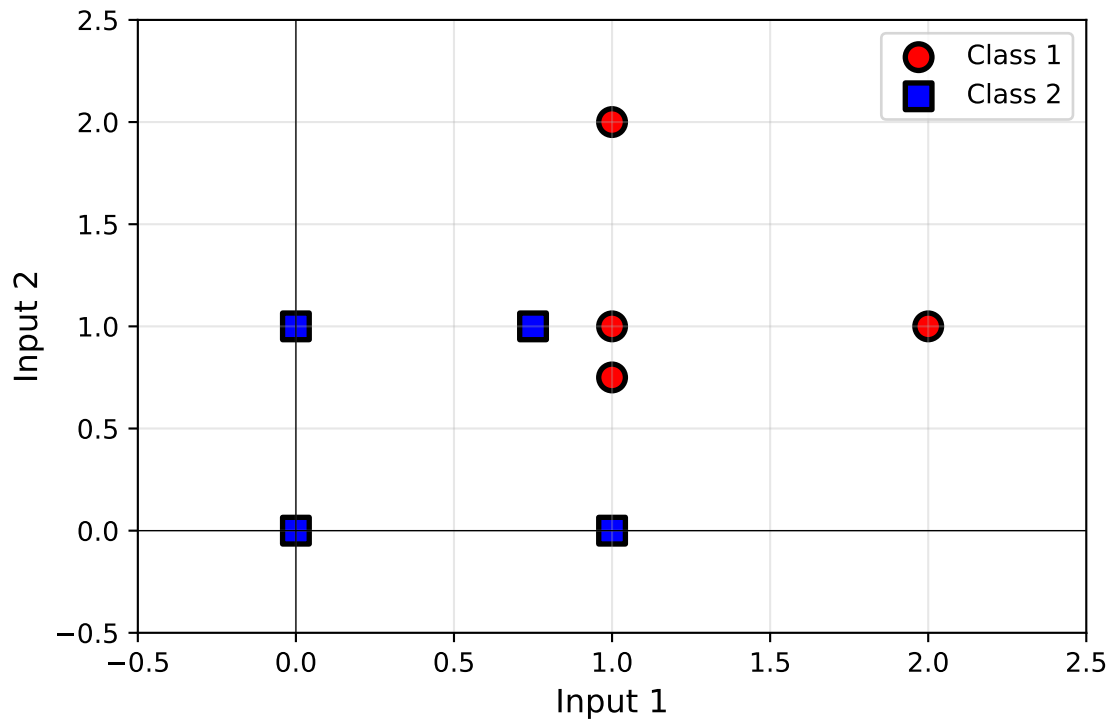
The margin is of size $\frac{2}{\sqrt{2^2+2^2}} = 0.7071067812$

Support vectors are the points that lie directly on the class hyperplane. In this case:

Class 1 has the point (1, 1)

Class 2 has the points (0, 1), (1, 0)

5



$$w^T x^7 + b \rightarrow 2(1) + 2(0.75) - 3 = 0.5$$

$$0.5 \times y^i \geq 1 - \xi_7 \rightarrow 0.5 \geq 1 - \xi_7$$

$$\xi_7 = 0.5$$

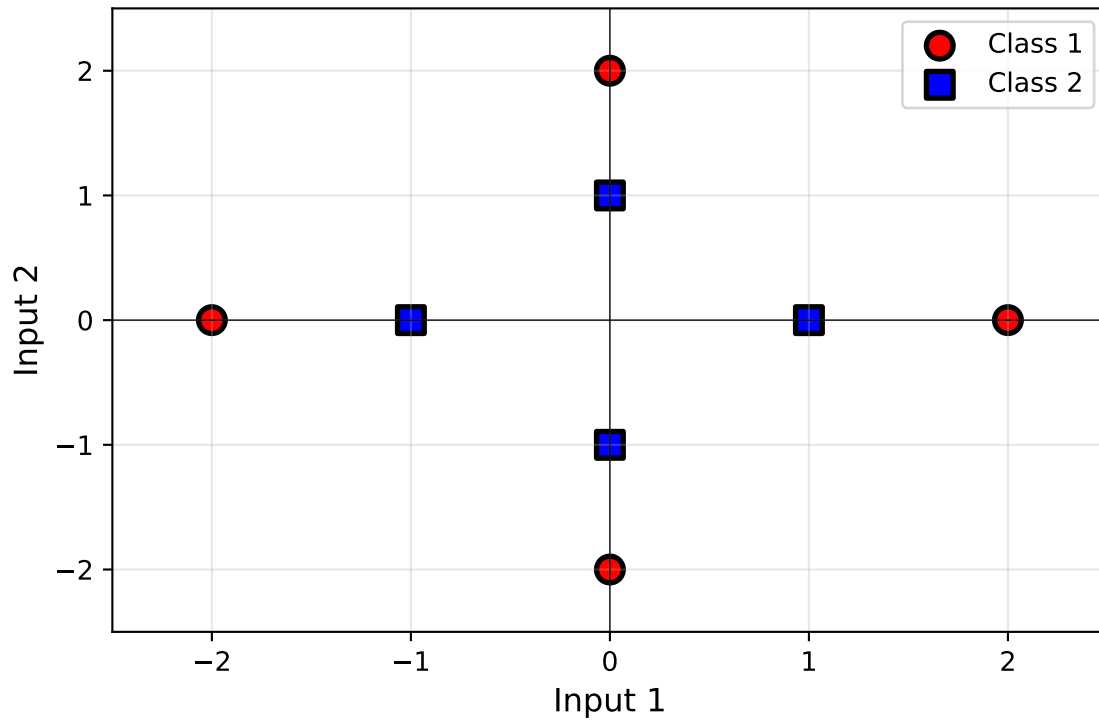
$$w^T x^8 + b \rightarrow 2(0.75) + 2(1) - 3 = 0.5$$

$$0.5 \times y^i \geq 1 - \xi_8 \rightarrow -0.5 \geq 1 - \xi_8$$

$$\xi_8 = 1.5$$

6

a



This is not linearly separable

b

$K(x^i, x^j) = (x^i \cdot x^j)^2 \rightarrow \phi(x^i) \cdot \phi(x^j)$ where $\phi(x) = (x_1^2, x_2^2, \sqrt{2}x_1x_2)$

When applying this to our current dataset:

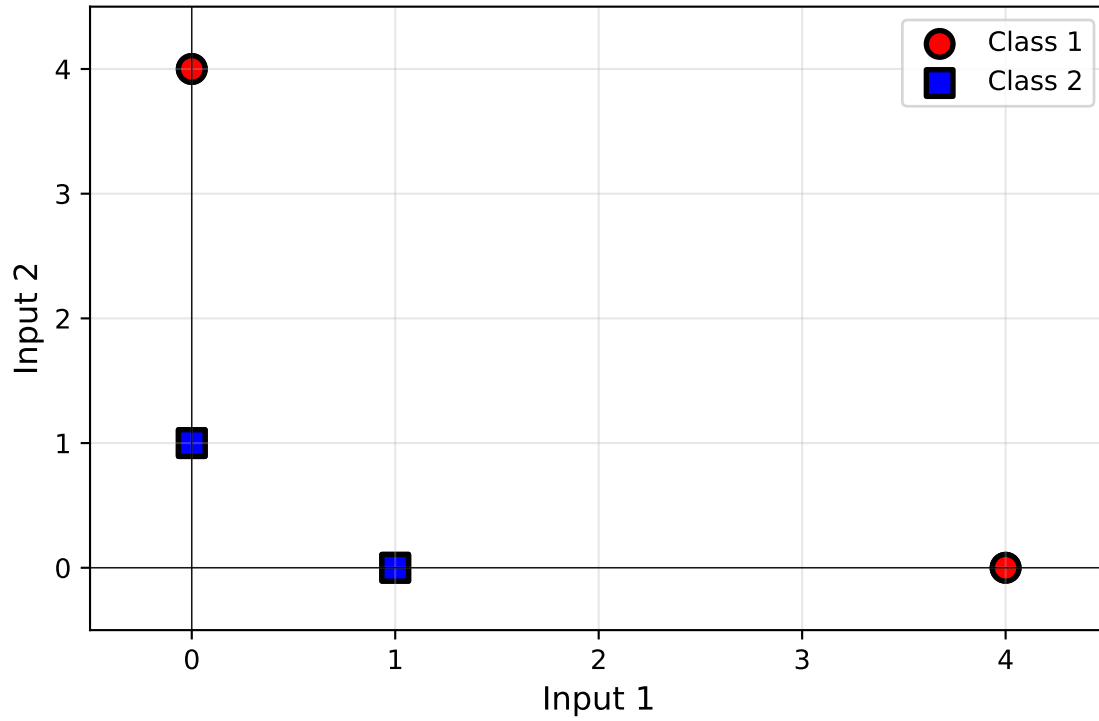
Class 1:

$$\phi(x^1) = \begin{pmatrix} 4 \\ 0 \\ 0 \end{pmatrix}, \phi(x^2) = \begin{pmatrix} 0 \\ 4 \\ 0 \end{pmatrix}, \phi(x^3) = \begin{pmatrix} 0 \\ 4 \\ 0 \end{pmatrix}, \phi(x^4) = \begin{pmatrix} 4 \\ 0 \\ 0 \end{pmatrix}$$

Class 2:

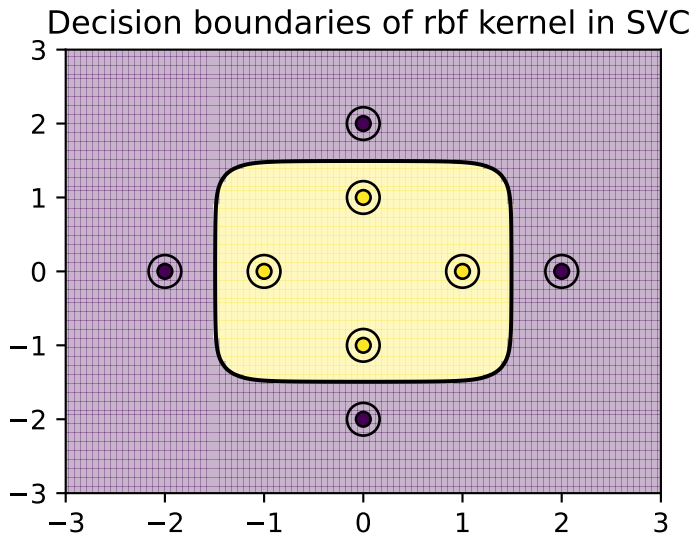
$$\phi(x^5) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \phi(x^6) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \phi(x^7) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \phi(x^8) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

Since all of the z values are 0 we can project our vectors onto a 2d plane as so



Here it is clear that this is linearly separable along the line $x + y - 2.5 = 0$

c



Gaussian kernel SVM can perfectly separate the two classes

7

The code submitted tests a variety of different kernels (linear, polynomial and RBF/Gaussian) with different regularization parameters (0.01, 0.1, 1, 10, 100), different gamma values (0.01, 0.1, 1, 10) and degree values (2, 3, 4) for the kernels. Slight caveat that gamma = 10 and $C = 100$ together takes too long and was skipped. This resulted in the best model being an RBF/Gaussian kernel with a regularization parameter of 100 and a gamma value of 0.01 which achieved an accuracy of 99.69% on the test set. With the best polynomial kernel being a degree 3 polynomial with a regularization parameter of 11 and a gamma value of 0.1 which achieved an accuracy of 99.45% on the test set. The best linear kernel was with a regularization parameter of 1 which achieved an accuracy of 99.48% on the test set.